ModelIt!

Student Activity Pack

Evolution in Action: How Species Change Over Time

Why Biodiversity Is Nature's Insurance Policy

Grade 07L2 | NGSS MS-LS4-1, MS-LS4-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think happens to a population of animals when their habitat suddenly changes?

2 Why do you think there are so many different species of the same type of animal (like hundreds of beetle species)?

3 Draw what you think happens to a population over many generations if only the fastest individuals survive.

4 Why do you think some species go extinct while others survive the same environmental change?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Environmental Change Rate | Population Size | Genetic Variation | Selection Pressure | Adaptation Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Rapid Change + Small Population

Fast environmental shift, few individuals

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Rapid Change + Large Population

Same fast change, many individuals

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Gradual Change

Slow environmental shift over many generations

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Geographic Isolation — When a physical barrier separates a population into groups that can no longer interbreed, leading to independent evolution

Mutation Rate — The frequency of new genetic changes arising during reproduction, providing fresh raw material for natural selection

Competition Intensity — The degree to which organisms compete for limited resources, influencing which traits provide advantages

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a population of identical-looking organisms become two completely different species over time? Use evidence from your simulation to support your answers.

1. **Why can't individual organisms evolve? Why does evolution only happen at the population level?**

2. **How does your model show that genetic variation is the raw material for natural selection?**

3. **What would happen in your model if the environment changed so fast that no individuals had suitable traits?**

4. **How does the fossil record provide evidence that species have changed over millions of years?**

5. **Why is maintaining biodiversity important for the future of life on Earth?**

8 STEM Challenge: Design a Conservation Strategy

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YOUR ENGINEERING MISSION

Create an evidence-based conservation strategy for an endangered species, using your evolution model to explain why genetic diversity is critical for the species' survival.

THE SCENARIO

A wildlife conservation organization has asked your team to develop a rescue plan for an endangered species with a shrinking population. Using your model data, you need to explain why the population is at risk and propose strategies to increase their chances of survival.

Engineering Design Thinking

1. WHICH ENDANGERED SPECIES WILL YOU FOCUS ON AND WHAT ENVIRONMENTAL PRESSURES DOES IT FACE?
2. HOW WILL YOU USE MODEL EVIDENCE TO JUSTIFY YOUR CONSERVATION RECOMMENDATIONS?
3. WHAT SPECIFIC STRATEGIES WILL INCREASE POPULATION SIZE AND GENETIC DIVERSITY?
4. HOW WILL YOU MEASURE WHETHER YOUR CONSERVATION STRATEGY IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation Biologists study ecosystems and develop strategies to protect endangered species and habitats. They work for wildlife agencies, national parks, and environmental nonprofits, earning $55,000-$95,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the evolution model, a small population with low genetic variation faces rapid environmental change. The model predicts extinction. Which chain of relationships in the model explains this outcome?

- **A.** Small Population Size leads to high Genetic Variation, which leads to fast Adaptation Rate, but the environment changes too quickly
- **B.** Small Population Size leads to low Genetic Variation, which limits Adaptation Rate because natural selection has few trait options to act on, and the population cannot keep pace with environmental change
- **C.** Environmental Change Rate directly reduces Population Size without involving Genetic Variation or Selection Pressure
- **D.** Small populations evolve faster because there are fewer individuals to change

2 A student claims that organisms evolve on purpose to survive environmental changes. Using the model, which response best corrects this misconception?

- **A.** Organisms do evolve on purpose, but only over many generations
- **B.** Evolution acts on random genetic variation that already exists in the population. Natural selection favors traits that happen to match the new conditions, not traits organisms chose to develop
- **C.** Individual organisms change their DNA during their lifetime to adapt to new conditions
- **D.** Evolution only occurs through mutations caused by environmental stress

3 The model shows that when Environmental Change Rate is set to gradual (low), even populations with moderate Genetic Variation can adapt successfully. Which principle explains why slow change is easier for populations to survive?

- **A.** Slow environmental change produces weaker Selection Pressure, giving natural selection more time to shift trait frequencies across generations before conditions become critical
- **B.** Slow environmental change does not create any Selection Pressure at all
- **C.** Populations can sense gradual changes and prepare their DNA accordingly
- **D.** Gradual change only affects individual organisms, not entire populations

4 Two populations of the same species live in separate environments. Population A is large with high genetic variation. Population B is small with low genetic variation. Both face the same disease outbreak. Based on the model, which prediction is most supported?

- **A.** Both populations will be equally affected because the disease does not distinguish between population sizes
- **B.** Population A is more likely to survive because its higher genetic variation means some individuals are more likely to carry disease resistance traits
- **C.** Population B will survive because smaller populations are easier to manage during disease outbreaks
- **D.** Neither population will be affected because natural selection prevents disease from spreading

5 **The fossil record shows that over 99% of all species that ever lived are now extinct. Based on the evolution model, which factor most strongly determines whether a species survives or goes extinct during a mass extinction event?**

- **A.** How large and physically strong the species is
- **B.** Whether the species has enough genetic variation to include individuals with traits suited to the dramatically changed post-extinction environment
- **C.** How old the species is, because older species have more experience surviving
- **D.** Whether the species actively chooses to migrate to safer habitats